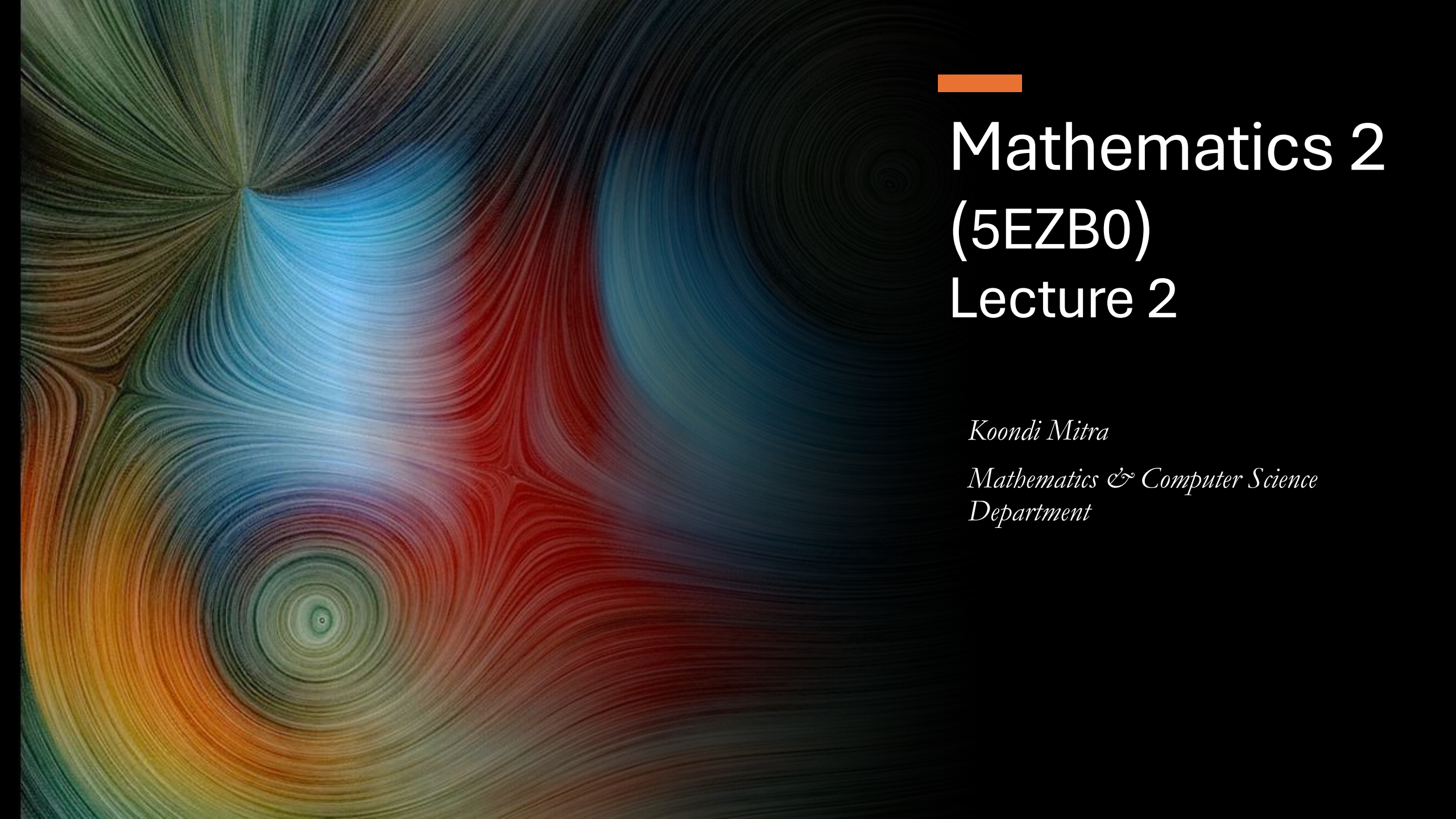




Mathematics 2 (5EZB0) Lecture 2

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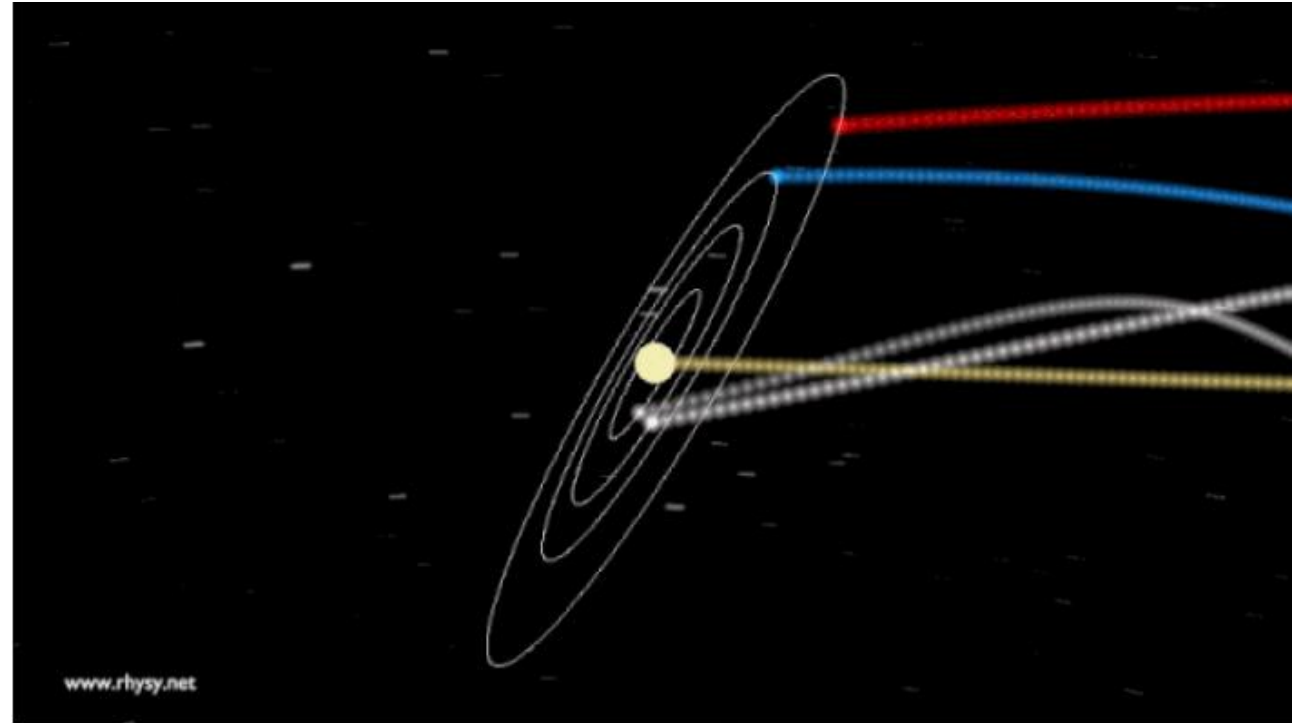
Vector functions of one variable

Motion is characterized by a position vector

$$\mathbf{r}(t) = x(t) \mathbf{i} + y(t) \mathbf{j} + z(t) \mathbf{k}$$

Changing with time $t \in \mathbb{R}$

Use **functions_of_time.py** to see how the position vector changes in 3D for any given function of time



Velocity

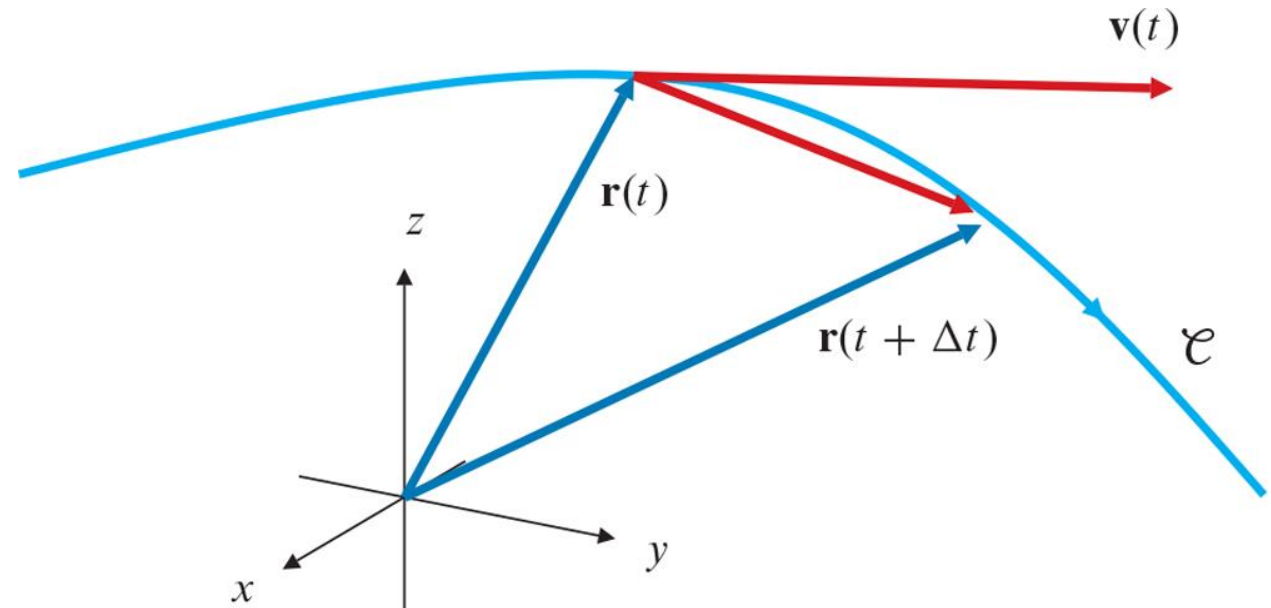
The instantaneous rate of change of the position vector is called *velocity*

$$\mathbf{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt}$$

- The velocity vector's components are velocities of the components

$$\frac{d\mathbf{r}}{dt} = \frac{dx}{dt} \mathbf{i} + \frac{dy}{dt} \mathbf{j} + \frac{dz}{dt} \mathbf{k}$$

- Speed: $v(t) = |\mathbf{v}(t)|$
- Velocity $\mathbf{v}(t)$ is tangent to the trajectory at the point $\mathbf{r}(t)$



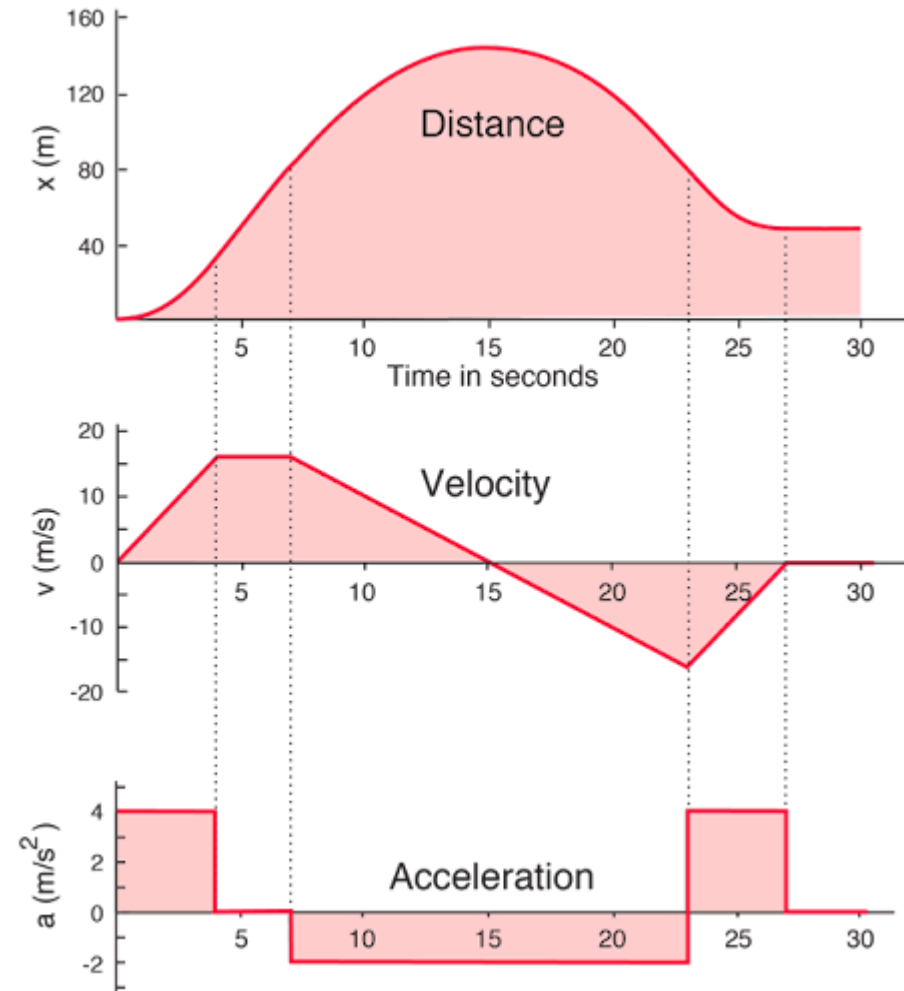
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Acceleration

The instantaneous rate of change of velocity is called *acceleration*:

$$\mathbf{a}(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{v}(t + \Delta t) - \mathbf{v}(t)}{\Delta t} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$$

➤ $\mathbf{a}(t) = \frac{d^2x}{dt^2}\mathbf{i} + \frac{d^2y}{dt^2}\mathbf{j} + \frac{d^2z}{dt^2}\mathbf{k}$



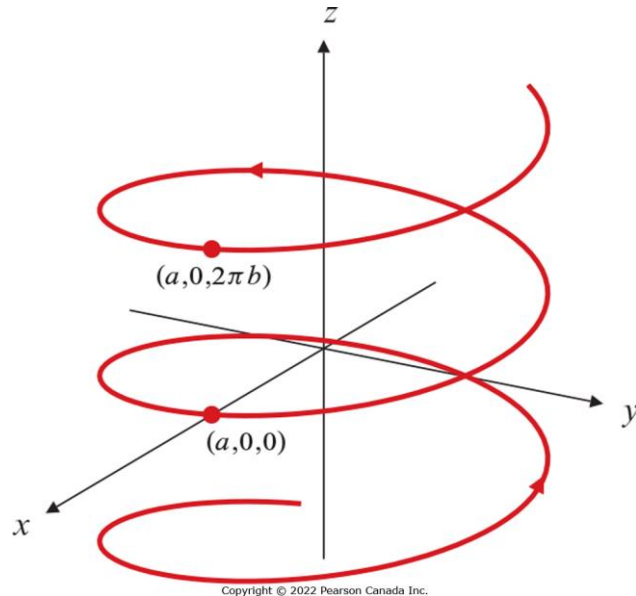
Velocity + Acceleration

velocity

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}$$

acceleration

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2} = \frac{d^2x}{dt^2}\mathbf{i} + \frac{d^2y}{dt^2}\mathbf{j} + \frac{d^2z}{dt^2}\mathbf{k}$$



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Questions 1

1. Conical helix: What is the velocity, speed, and acceleration of the particle with coordinate vectors $(t \cos t, t \sin t, t)$?
2. What is the position vector and velocity of a particle having constant acceleration $\mathbf{g}$?
3. An ant walks around a cylinder of radius 2 cm oriented along z-axis, with constant speed 5 cm/s, and goes along the axis with speed 3 cm/s. If it starts $(2, 0, 0)$, what is its velocity and speed at that point?

Rules of differentiation

- $\frac{d}{dt}(\mathbf{u}(t) + \mathbf{v}(t)) = \frac{d}{dt}\mathbf{u}(t) + \frac{d}{dt}\mathbf{v}(t)$ (Addition rule)
- $\frac{d}{dt}(\lambda(t) \mathbf{u}(t)) = \frac{d}{dt}\lambda(t) \mathbf{u}(t) + \lambda(t) \frac{d}{dt}\mathbf{u}(t)$ (Product rule)
- $\frac{d}{dt}(\mathbf{u}(t) \cdot \mathbf{v}(t)) = \frac{d}{dt}\mathbf{v}(t) \cdot \mathbf{u}(t) + \mathbf{v}(t) \cdot \frac{d}{dt}\mathbf{u}(t)$ (Product rule for dot products)
- $\frac{d}{dt}(\mathbf{u}(t) \times \mathbf{v}(t)) = \mathbf{u}(t) \times \frac{d}{dt}\mathbf{v}(t) + \frac{d}{dt}\mathbf{u}(t) \times \mathbf{v}(t)$ (Product rule for cross products)
- $\frac{d}{dt}(\mathbf{u}(\lambda(t))) = \frac{d}{d\lambda}\mathbf{u}(\lambda(t)) \frac{d}{dt}\lambda(t)$ (Chain rule)
- $\frac{d}{dt}|\mathbf{u}(t)|^s = s|\mathbf{u}(t)|^{s-2} \mathbf{u}(t) \cdot \frac{d}{dt}\mathbf{u}(t)$

Questions 2

1. Show that the speed of particle remains constant only if the acceleration and velocity are perpendicular.

2. What is $\frac{d}{dt}\left(\mathbf{r} \times \frac{d}{dt}\mathbf{r}\right)$? (an important relation)

Parametrization

Time can be used as a parameter to describe a surface, curve: for $a \leq t \leq b$,

$$\mathbf{r}(t) = x(t) \mathbf{i} + y(t) \mathbf{j} + z(t) \mathbf{k}$$

- Parametrization is not necessarily unique
- There can be more than 1 parameter (if you need to describe a surface for example)

Example

1. $\mathbf{r}(t) = \sin t \mathbf{i} + \cos t \mathbf{j}, -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$
 2. $\mathbf{r}(t) = t\sqrt{2-t^2} \mathbf{i} + (1-t^2) \mathbf{j}, -1 \leq t \leq 1$
- are both parametrizations of the unit circle
- $$x^2 + y^2 = 1$$

Question 3

The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ can be parametrized as $(a \cos t, b \sin t)$ for $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$. Propose a parametrization for the ellipsoid

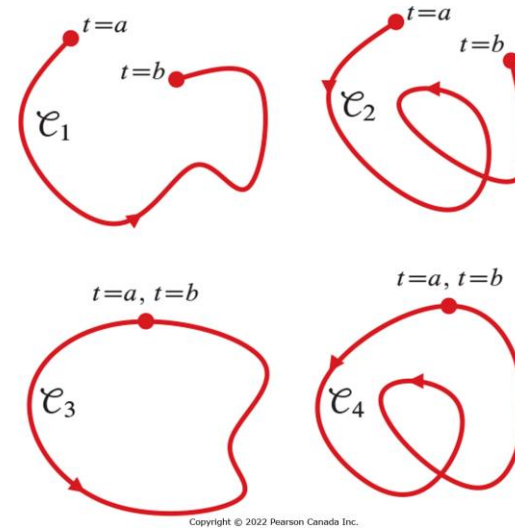
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Parametrizing

- Time can be used as a parameter to describe a surface, curve: for $a \leq t \leq b$,

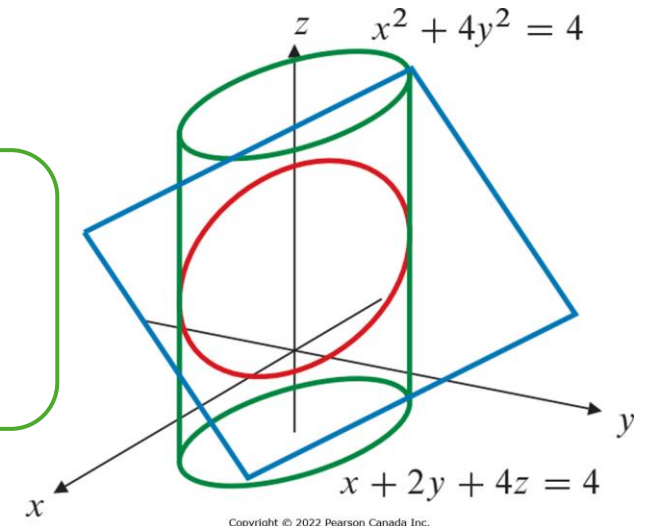
$$\mathbf{r}(t) = x(t) \mathbf{i} + y(t) \mathbf{j} + z(t) \mathbf{k}$$

- Parametrization is not necessarily unique
- There can be more than 1 parameter (if you need to describe a surface for example)
- Can be **closed** ($\mathbf{r}(a) = \mathbf{r}(b)$) or **open** curves
- Can even be self intersecting



Question 4

Find a parametrization of the red curve on the right.



Arc length

- Arc length for the curve $\mathbf{r}(t)$ where $a \leq t \leq b$

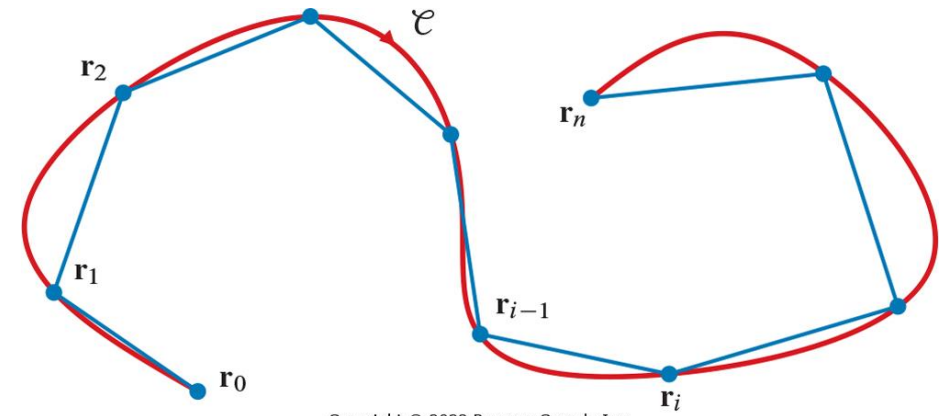
$$s = \int_a^b \left| \frac{d\mathbf{r}}{dt} \right| dt = \int_a^b v(t) dt$$

- The arc length is independent of the parametrization
- The length of the curve $\mathbf{r}(x) = x \mathbf{i} + f(x) \mathbf{j}$ for $a \leq x \leq b$,

$$s = \int_a^b \sqrt{1 + f'(x)^2} dx$$

- The length of the curve (polar coordinates) $r = g(\theta)$ for $a \leq \theta \leq b$,

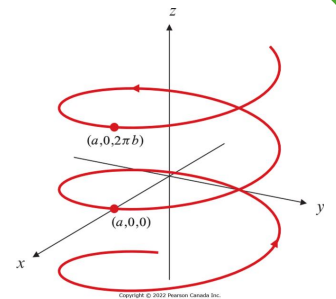
$$s = \int_a^b \sqrt{g(\theta)^2 + g'(\theta)^2} d\theta$$



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Questions 5

What is the length the ant travels while going around the cylinder once.



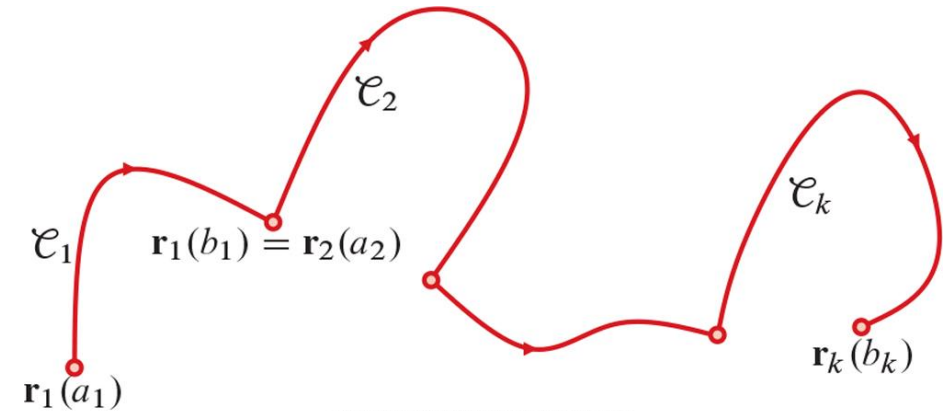
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Arc length parametrization

- **Piecewise smooth curves:** if the curve is piece-wise smooth then take the piece-wise sum or arc-lengths
- **Arc length parametrization:** The arc length between two point is parameter independent. Hence, the most natural (intrinsic) parametrization is when you parametrize the curve based on the arc length s . For a curve $\mathbf{r}(t)$, let

$$s(t, 0) = L(t)$$

Then $\mathbf{r}(s) = \mathbf{r}(L^{-1}(s))$ is its intrinsic parametrization



Questions 6

1. Parametrize the ant's position in terms of the arc length.
2. What is arc length parametrization corresponding to Question 1.1?